Roll No. 0005

CSET485 – AI and Society

Assignment 1: Stress-Testing a Predictive Model

Dataset	German Credit Dataset (UCI Statlog, id 144)
Random Seed	5

This report covers the German Credit data. My roll number is 0005, so I was assigned the dataset for last digits 3, 4 and 5, and my random_state is 5. Every number below comes from my notebook (CSET485_Assignment1_Roll0005.ipynb), and the few values I worked out by hand from its outputs are marked as derived. The target is approval (1 = good credit, 0 = bad credit), so the flagged group is the one with a score at or above the threshold, which means the people the model predicts will be approved. The assignment calls that group high risk because it was written around recidivism scores, but here the flagged group is the high-score, likely-to-be-approved one.

Two results need special handling, so I want to say so up front. The search for Simpson's paradox in Section 3.1 did not find a statistically significant reversal (the credit-amount sign flip has confidence intervals that all include zero), and the 0.05 threshold in the fairness sweep in Section 3.4 is a trivial test because it approves everyone.

Code: https://github.com/Atindriya27/Stress-Testing-a-Predictive-Model-

1. Data Preprocessing

- ### Dataset and Target

  The file has 1,000 loan applicants, 20 attributes and one class label: 700 good credit and 300 bad credit. I loaded it with the ucimlrepo package and renamed the columns using the official attribute definitions. The UCI file codes good credit as 1 and bad as 2, so I recoded it to approved = 1 (good credit) and 0 (bad credit). A high score therefore means an applicant the model likes.

- ### Missing Values

  There are no missing values. The null count is 0 in all 20 features, so no rows were dropped. My pipeline still includes a safeguard imputer (median for numeric, most-frequent for categorical), but it changed nothing here.

- ### Categorical Encoding

  I one-hot encoded the 13 categorical attributes (checking_status, credit_history, purpose, savings_status, employment, personal_status, other_debtors, property, other_installment_plans, housing, job, telephone, foreign_worker) and set the encoder to ignore categories it hasn't seen. With the 7 numeric columns, that gives 61 features (700 × 61 for training and 300 × 61 for testing).

- ### Outliers

  I kept every row. Large credit amounts and applicants as old as 74 in the test set are real people, and removing them would hide exactly the cases a lender meets. Standardising the numeric columns and using a regularised logistic fit limits how much pull they have. This was a deliberate choice, not something I overlooked.

- Scaling and split

  I split the data first and scaled afterwards. With random_state = 5 and a 70/30 split stratified on the target, I got 700 training rows and 300 test rows. The 7 numeric columns (duration, credit_amount, installment_rate, residence_since, age, existing_credits, dependents) were standardised with a StandardScaler fitted on the training rows only, so no test information leaks into the model. The dummy variables stay as 0/1.

A note when comparing tables: the personal-status split in Section 3.1 uses the original, unscaled credit_amount, age and duration, while the age split uses the 61 standardised features, so the same variable has different coefficients in different places.

2. Part 1: Build and compare

Model A is an ordinary least squares regression on the 0/1 approval label, and I cut its test predictions at their own median, **0.6835**. Model B is a logistic regression (max_iter = 1000, default L2 penalty), and I cut its probabilities at 0.5. Both use the same 700 training rows and 61 features and are scored on the same 300 test rows. Here is how the logistic model does on the test set:

Table 1. Logistic regression performance on the 300 test cases.

Accuracy	Precision	Recall	ROC-AUC	Recall, class 0 (bad credit)	Recall, class 1 (good credit)
0.7367	0.7860	0.8571	0.7952	0.46	0.86

Agreement between the two thresholding approaches (rows = linear median rule, columns = logistic 0.5 rule):

Table 2. Agreement between the median-threshold OLS rule and the 0.5-threshold logistic rule (rows = OLS, columns = logistic).

	Logistic = 0	Logistic = 1	Total
Linear = 0 (below median)	71	79	150
Linear = 1 (≥ median)	0	150	150
Total	71	229	300

The two rules disagree on **79 of 300 applicants (26.33%)** and agree on 221 (73.67%). All 79 disagreements are people the linear rule puts in the lower half but the logistic model approves. There are none in the other direction.

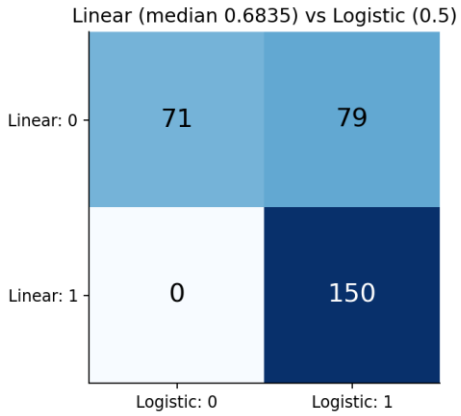


Figure 1. Confusion matrix of the two classification rules.

Why they differ. The median of the linear predictions (0.6835) sits well above 0.5, and a median rule always splits people 50/50 while the data are about 70% good credit. Question 1 in Part 3 goes through this in full.

3. Part 2: The trap

- 3.1 Simpson's Paradox hunt

I tried two splits. **Split 1 is personal status.** Group A is A91/A92 (246 training cases, 160 with good credit = 65.04%) and Group B is A93/A94 (454 cases, 330 with good credit = 72.69%). By individual status the good-credit rates are A91 55.88%, A92 66.51%, A93 72.05% and A94 76.56%. The predictor I look at is credit_amount (logistic regression on credit_amount + age + duration).

Table 3. credit_amount coefficient by personal-status group (logistic regression on credit_amount + age + duration).

Model	n	credit_amount coef.	95% CI	p-value	Odds ratio per DM 1,000
Full training set	700	-2.889e-05	[-1.02e-04, 4.4e-05]	0.440	0.972
Group A (A91/A92)	246	+5.1e-05	[-8.8e-05, 1.91e-04]	0.468	1.052
Group B (A93/A94)	454	-7.0e-05	[-1.57e-04, 1.7e-05]	0.114	0.932

The credit_amount coefficient does **change sign** (positive for Group A, negative for Group B), but all three intervals include zero. A Wald-type test on the A-versus-B difference (1.21e-04), with standard errors recovered from the two intervals, gives z = 1.44 and p ≈ 0.149, so it isn't statistically significant. **Split 2 is age** (under 30 against 30 and over; 256 against 444 training cases, with good-credit rates of 63.3% and 73.9%), using all 61 features:

Table 4. Age coefficient by age group (all 61 features; unregularised statsmodels fit and regularised refit).

Model	Age coef. (statsmodels Logit)	95% CI	p-value	Age coef. (regularised, C = 1)
Full dataset	0.1310	[-0.1105, 0.3724]	0.288	0.1219
Younger (<30)	1.3425	[-0.4821, 3.1671]	0.149	0.4271
Older (≥30)	0.0012	[-0.3453, 0.3478]	0.994	0.0050

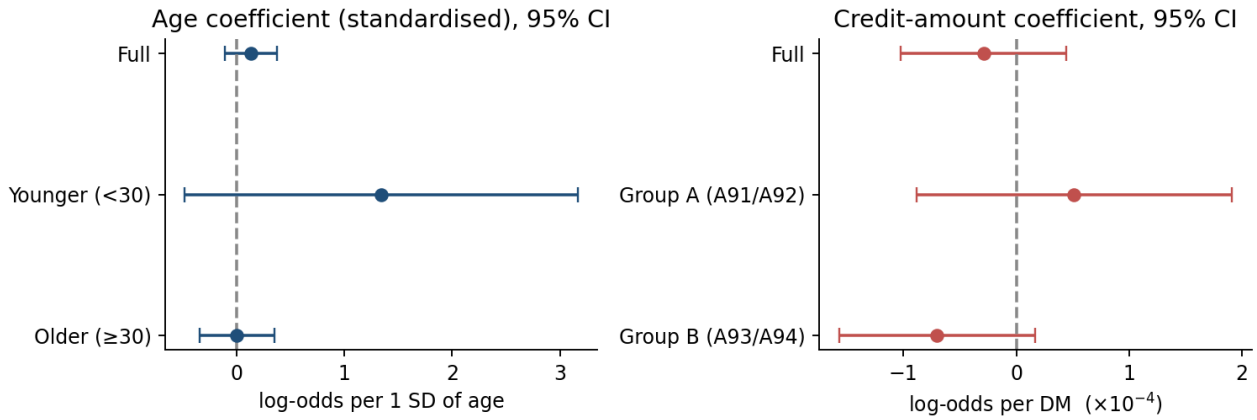


Figure 2. Subgroup coefficients with 95% confidence intervals (dashed line = no effect).

A direct interaction test (approval ~ age_centered × age_group) gives a coefficient of 0.0446 (95% CI -0.0557 to 0.1450, p = 0.384), which again isn't significant. The unregularised model for under-30s also threw a ConvergenceWarning (61 features on 256 rows), which is why its coefficient looks inflated, so the regularised value of 0.4271 is more believable. **Overall,** I found no statistically

significant reversal. The sign flip in credit amount and the stronger age effect among under-30s are suggestive, but they could easily be noise. Question 2 in Part 3 explains this and the policy implication.

- ## 3.2 Omitted Variable bias

I dropped checking_status. Full model: logistic regression of approval on all 20 original predictors (training set, n = 700). Reduced model: the same 19 predictors without checking_status. Both models converged.

Table 5. Coefficients with and without checking_status. % change = |reduced − full| / |full|.

Predictor	Full coefficient	Reduced coefficient	Absolute change	% change
credit_amount	-0.000146	-0.000144	0.000002	1.65%
age	0.011734	0.014400	0.002665	**22.71%**
duration	-0.022638	-0.025317	0.002678	11.83%
installment_rate	-0.377599	-0.373549	0.004050	1.07%

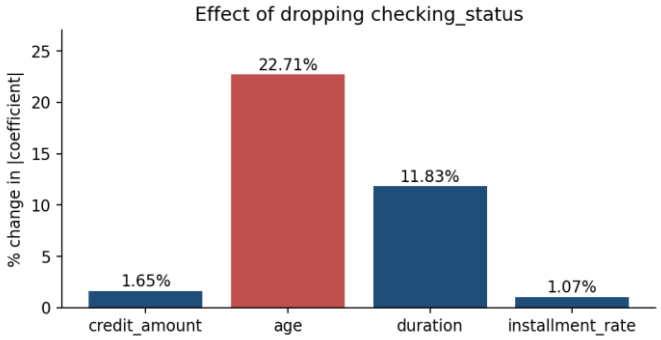


Figure 3. Percentage shift in each coefficient after dropping checking_status.

Age shifted the most (22.71%), followed by duration (11.83%), credit_amount (1.65%) and installment_rate (1.07%). None of the coefficients changed sign. Questions 3 and 4 in Part 3 discuss this and how to detect hidden confounds.

- ## 3.3 Adversarial subset

On the 300 test cases I looked for confident mistakes: P(approve) ≥ 0.75 where the actual outcome was 0 (false positive), or P ≤ 0.25 where it was 1 (false negative). Only **20 cases (6.67% of the test set)** qualify, so I used all of them instead of sampling 50. Seventeen (**85%**) are false positives and three are false negatives (15%).

Table 6. The 20 high-confidence errors compared with the full test set.

Characteristic	Adversarial subset (n = 20)	Full test set (n = 300)
Mean age	29.65	35.93
Median age	28	33
75th percentile of age	35.25	42
Maximum age	43	74
Mean credit amount (DM)	3,244.05	3,370.73
Mean duration (months)	21.15	20.97
Personal status A93 / A92 / A94 / A91	40% / 35% / 25% / 0%	52.67% / 32.67% / 9.33% / 5.33%

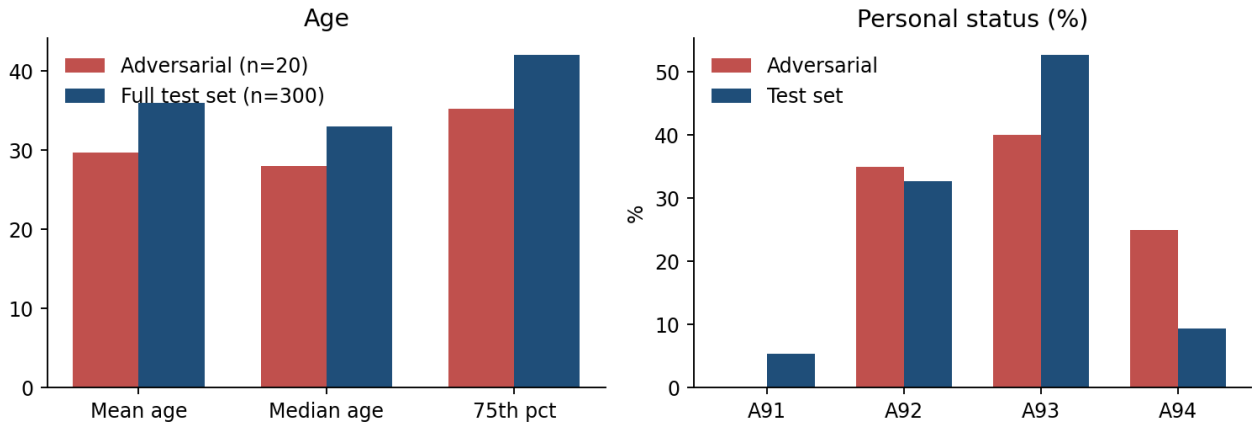


Figure 4. Adversarial subset versus full test set.

What they share is youth. The mean age is 6.3 years lower than in the test set, nobody is older than 43, and 75% (15 of 20, inferred from the 75th percentile) are 35 or younger. In the ten rows my notebook printed, 8 had the lowest savings category (A61) and 5 had no checking account (A14), but I didn't tabulate this for all 20, so I treat it as a hint only. The statistical explanation is in Question 5.

- ## 3.4 Fairness tradeoff analysis

I used personal status as the sensitive variable. Group A is A91/A92 (114 test applicants) and Group B is A93/A94 (186). I swept the logistic threshold from 0.05 to 0.95 in steps of 0.10. DP = $P(\hat{Y}=1|A)/P(\hat{Y}=1|B)$; EO = $\min(TPR_A/TPR_B, FPR_A/FPR_B)$; PP = PPV_A/PPV_B.

Table 7. Fairness metrics across the threshold sweep (highlighted row = the threshold I recommend). Ideal value for DP, EO and PP is 1.0.

Threshold	DP	EO	PP	Accuracy
0.05	1.0000	1.0000	0.8334	0.7000
0.15	0.9896	0.9937	0.8421	0.7200
0.25	0.9624	0.9789	0.8476	0.7367
0.35	0.9172	0.9204	0.8364	0.7333
0.45	0.8571	0.8978	0.8786	0.7633
0.55	0.8215	0.7742	0.9180	0.7433
0.65	0.7956	0.7287	0.9480	0.7333
0.75	0.6746	0.4554	0.9990	0.6767
0.85	0.5370	0.3123	1.0128	0.5900
0.95	0.4737	0.5684	1.0000	0.4333

The highlighted row (t = 0.45) is the threshold I recommend. At t = 0.05 everyone is approved, so DP = EO = 1.0 is trivial (accuracy equals the 0.70 base rate). PP above 1 at t = 0.85 and the rebound of EO at t = 0.95 come from very few approvals and small samples.

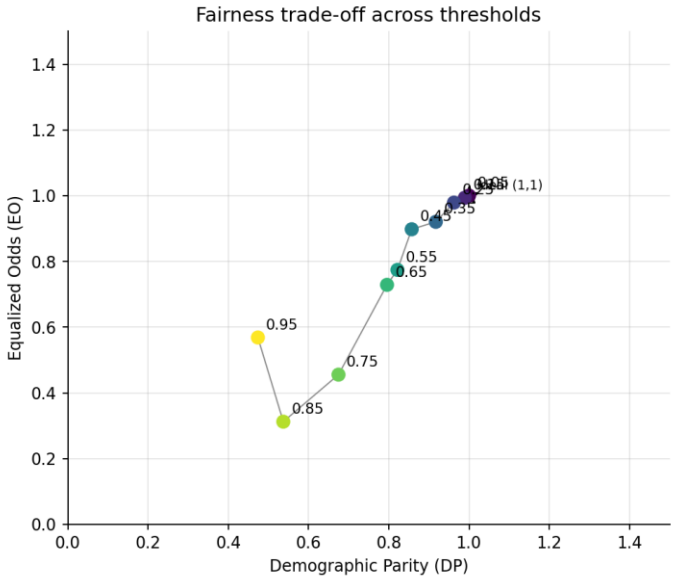


Figure 5. Trade-off chart: DP (x) against EO (y), each point labelled with its threshold; the star marks the ideal (1, 1).

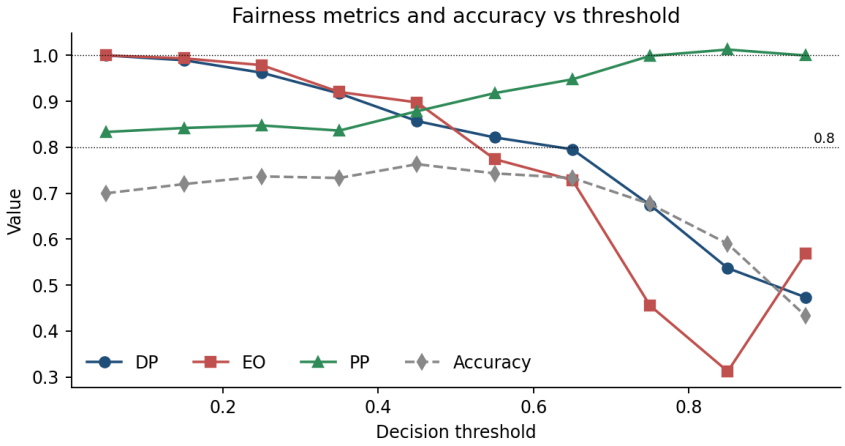


Figure 6. DP, EO, PP and accuracy as functions of the threshold.

Quantifying the trade-off (percentages are from the notebook; the PP gain and cost ratios are simple arithmetic on the table). (a) Lowering t from 0.75 to 0.45 improves DP from 0.6746 to 0.8571 (+27.05%) and EO from 0.4554 to 0.8978 (+97.14%), but PP falls from 0.9990 to 0.8786 (−12.05%), so the cost is 12.05 / 27.05 = **0.45** of PP lost per unit of DP gained. (b) Going the other way, raising t from 0.45 to 0.75 gains 13.70% in PP but loses 49.28% in EO and 21.29% in DP, a cost of 49.28 / 13.70 = **3.60**. **Key insight:** DP and EO move together here while PP moves the opposite way, so no threshold satisfies all three. PP is highest (0.9990) exactly where DP and EO are at their worst (0.6746, 0.4554). I recommend t = 0.45, and Questions 7 and 8 argue for it.

4. Part 3: Descriptive analysis questions

- Question 1

  Explain in detail why the threshold-based approach and the direct logistic regression approach did, or did not, flag the same individuals as "high risk" in your data. Quote the specific count/percentage of mismatched cases. Reference the median predicted value from linear regression and how it compares to the 0.5 threshold in logistic regression.

- Out of 300 test applicants, the two rules disagreed on **79 (26.33%)** and agreed on 221 (73.67%). The disagreement only goes one way. In the confusion matrix, 79 people fall below the median under the linear rule but get approved by the logistic model, and nobody sits in the opposite cell. So the real question is why the two cut-offs are so far apart. The median of the linear model's test predictions is **0.6835**, while logistic regression uses **0.5**. A median rule always flags exactly half of the people, which here is 150 of 300. Logistic at 0.5 flags 229 (76.3%), which is much closer to reality, since 210 of the 300 test applicants (70%) actually have good credit. Anyone scoring between about 0.5 and 0.6835 falls into the gap, and that gap is the 79 mismatches. Test case 2 is a good example: its linear score is 0.533 and its logistic probability is 0.548, so logistic approves while the median rule doesn't. All 150 people flagged by the linear rule are also flagged by logistic, which tells me the two models rank applicants in nearly the same order. They just draw the line in different places. There is a smaller second reason as well. OLS fits a straight line to a 0/1 outcome, so its scores aren't real probabilities (case 5 gets 0.266 from OLS and 0.224 from logistic), while the logistic curve pushes scores toward 0 and 1. Overall I think the 0.5 logistic rule is the better choice here, because forcing half the applicants into the 'low' group contradicts a 70/30 split. Its test accuracy is 0.7367 (precision 0.786, recall 0.8571, ROC-AUC 0.7952), although it only catches 46% of the bad-credit applicants.

- Question 2

  Describe the subgroup you found (or the subgroup with significantly different effects). Using your actual regression coefficients from the full model, subgroup A, and subgroup B, explain why the trend reverses or differs. Cite the exact coefficient values and confidence intervals. Explain what policy conclusion would change if the subgroup were ignored.

- I tried two splits, and neither gives a statistically significant reversal, so I want to be upfront about that. The first split is by personal status. Group A (A91/A92) has 246 training cases and a 65.04% good-credit rate, and Group B (A93/A94) has 454 cases and 72.69%. For credit_amount, the full-model coefficient is **-2.889e-05** (95% CI -1.02e-04 to 4.4e-05, p = 0.440). For Group A it is **+5.1e-05** (CI -8.8e-05 to 1.91e-04, p = 0.468) and for Group B it is **-7.0e-05** (CI -1.57e-04 to 1.7e-05, p = 0.114). The sign flips, but every interval contains zero, and a Wald test on the A-versus-B difference gives z = 1.44, p = 0.149. That is suggestive at best. The second split is by age. The age coefficient is 0.1310 in the full model (CI -0.1105 to 0.3724), 1.3425 for under-30s (CI -0.4821 to 3.1671) and 0.0012 for people 30 and over (CI -0.3453 to 0.3478). The regularised version shows the same pattern (0.1219, 0.4271, 0.0050), but the interaction term is 0.0446 (CI -0.0557 to 0.1450, p = 0.384), so I can't call it real. The unregularised under-30 model also had a convergence warning, with 61 features on only 256 rows, which explains why 1.3425 looks so large. My best explanation is small subgroups and noise, plus the pooled model averaging two different slopes. If we ignored the subgroups, we would end up with a single rule, that bigger loans lower approval chances (odds ratio 0.972 per DM 1,000), applied to everyone, even though Group A's estimate points the other way. Since none of this is significant, the sensible policy conclusion is to collect more data before changing anything.

- Question 3

  State which variable you dropped for the omitted-variable test. Report the specific before/after shift in coefficients for at least 4 predictors—include a table. Calculate the percentage change for each.

Which coefficient shifted the most? What does this reveal about how much you can trust a model you did not stress-test this way?

- I dropped **checking_status**, which records the status of the applicant's current account. It is a strong predictor of credit quality and is probably related to loan size, duration and age. The full model is a logistic regression on **all 20 original predictors** (700 training cases), and the reduced model uses the other 19. Both converged. Percentage change is |reduced - full| / |full| x 100 (table below).

Predictor	Full	Reduced (no checking_status)	% change
credit_amount	-0.000146	-0.000144	1.65%
age	0.011734	0.014400	22.71%
duration	-0.022638	-0.025317	11.83%
installment_rate	-0.377599	-0.373549	1.07%

Age moved the most. Its coefficient went from 0.011734 to 0.014400 (+0.002665), a 22.71% change. Duration came next (-0.022638 to -0.025317, 11.83%), then credit_amount (-0.000146 to -0.000144, 1.65%) and installment_rate (-0.377599 to -0.373549, 1.07%). No coefficient changed sign. What I find interesting is that credit_amount and installment_rate hardly noticed, but once checking_status was gone, age looked almost a quarter more important and longer loans looked about 12% more harmful. That is what omitted-variable bias looks like: the missing variable is correlated with the ones that stay in, so they soak up part of its effect. Someone using the reduced model would overstate how much age matters, and nothing in its output would hint at it, because it converged fine and looks just as precise. In this case no conclusion flipped, but I could only run the test because I knew checking_status was in the data and chose to drop it. The worrying case is a confound I can't observe, which would distort coefficients in exactly the same hidden way. So I would trust a model that hasn't been stress-tested like this for ranking applicants (ROC-AUC 0.7952), but not for explaining why a coefficient has its value or for setting policy. Percentage changes also depend on a coefficient's size and units, so a small percentage on a large coefficient can still matter.

- Question 4

As a practitioner, how would you detect a hidden confound in your model that you cannot currently see? Propose one concrete, real-world method (e.g., domain expert audit, external validation on a different dataset, residual analysis, instrumental variable approach). This question has no clean answer; you will be graded on the realism and concreteness of your proposal.

- A hidden confound is a variable that sits behind both the inputs and the outcome but isn't in the data, so you can't see it in the coefficients. My proposal is a three-step audit that combines expert judgement with fresh data. Step 1: sit down with three to five loan officers and show them the 20 confident mistakes from my test set (17 were false positives, and the average age was 29.65 against 35.93 overall). Ask what they would want to know about these applicants that the form doesn't capture, such as how stable the employer is, any informal debts, or whether family would help out. Step 2: turn their answers into measurable variables and collect them for a fresh sample of around 500 loans from another branch or a later year. Step 3: test each variable in two ways. First, check whether it is correlated with my model's residuals. Second, add it to the model and see whether the existing coefficients move. My own omitted-variable test shows what to look for, since dropping checking_status shifted the age coefficient by 22.71% and duration by 11.83%. I would agree on a rule in advance, for example that a shift above 10% in a key coefficient, or a likelihood-ratio test with $p < 0.05$, means the model needs revising. I would also run the unchanged model on the new sample. Its test ROC-AUC is 0.7952, so a drop to around 0.72 would point to a missing driver or a shift in the applicant population. Residual plots by age band and personal status would show where

the errors pile up. This is realistic and fairly cheap, but it can't prove that nothing is hiding. It only lowers the chance that I'm missing something large.

- Question 5

Describe your 50-row adversarial subset. What common characteristic do these rows share, and what percentage of the subset exhibits this trait? Statistically, why does this characteristic cause your model to fail confidently? Reference your model's coefficients or learned decision boundaries to explain the failure mechanism.

- Only **20** test cases (6.67% of 300) were confidently wrong, so I used all of them instead of sampling 50. Seventeen of them (**85%**) are false positives, where the model gave a probability of at least 0.75 and the person actually had bad credit, and three are false negatives. What they share is that they are young. The average age is 29.65 against 35.93 in the test set, the median is 28 against 33, and the 75th percentile is 35.25 against 42. Nobody in the subset is older than 43, while the test set goes up to 74, and 75% of them are 35 or younger. Personal status A94 is also over-represented (25% against 9.33%), and A91 doesn't appear at all (0% against 5.33%). Loan size and length look normal (an average of 3,244 DM against 3,371 DM, and 21.15 months against 20.97), so the problem is who the applicants are, not how big the loan is. In the ten rows my notebook printed, 8 had the lowest savings category (A61) and 5 had no checking account (A14). I didn't count these across all 20, so I treat them as hints only. As for why the model fails so confidently, it is additive and linear in the log-odds, and it applies one age slope to everybody (0.1219 per SD in the regularised full model). The under-30 slope is 0.4271 and the over-30 slope is 0.0050, so a single slope can't describe both groups. With no interaction between age and the other strong signals, a young applicant who looks good on paper gets a high probability that the data doesn't really support. I would call this a missing interaction combined with small samples, not an optimisation problem.

- Question 6

Using your own fairness trade-off chart from Part 2.4, explain in concrete terms what the trade-off cost means for a real person affected by this decision. Describe a specific, realistic scenario: two individuals (one from each demographic group) and their outcomes under different decision thresholds. Show how prioritizing one fairness metric harms another group.

- Picture two applicants who would both repay a loan: Maya from Group A (a woman who is divorced or married, status A92) and Daniel from Group B (a single man, status A93). In my test set Group A has 114 people and Group B has 186. At t = 0.45, DP is 0.8571 and EO is 0.8978, so Maya is approved at roughly 86% of Daniel's rate and the error rates are within about 10% of each other. Now imagine a cautious lender raises the threshold to 0.75 to be surer about the loans it approves. Predictive parity improves from 0.8786 to 0.9990 (+13.70%), which means approved people from both groups now default at almost the same rate. That looks fair from the lender's side. But DP drops to 0.6746 (-21.29%) and EO to 0.4554 (-49.28%). Group A's overall approval rate is only about 67% of Group B's, and among qualified applicants the ratio can be as low as 0.46. For every ten qualified people approved in Daniel's group, only four or five comparable people get approved in Maya's. Accuracy also falls from 0.7633 to 0.6767. The trade-off cost is 49.28 / 13.70 = 3.60 units of equalized odds lost for every unit of predictive parity gained. The opposite problem shows up at very low thresholds. At t = 0.05 everyone is approved, so DP and EO are exactly 1.0 and accuracy is just the 0.70 base rate. PP is 0.8334 there, meaning approved Group A applicants default more often than approved Group B ones. Maya might get a loan she can't sustain, along with the debt and credit damage that follows. So whichever metric we prioritise, someone carries the risk.

- Question 7

Choose a specific decision threshold from your sweep (e.g., t=0.35). Argue, using at least 3 specific numbers from your analysis (DP, EO, PP, accuracy, subgroup-specific rates), why this threshold is

defensible from a fairness standpoint. Describe the strongest counter-argument a critic could raise against your choice and acknowledge its validity.

- I would pick **t = 0.45**. At that threshold DP is 0.8571, EO is 0.8978 and PP is 0.8786, so all three are above 0.8, a common 'four-fifths' rule of thumb. Accuracy is 0.7633, the highest in the whole sweep and 6.3 points above the 0.70 you would get by approving everyone. It is also the highest threshold where all three metrics stay above 0.8. At t = 0.55, EO falls to 0.7742, and at t = 0.75 it drops to 0.4554 while accuracy slips to 0.6767. Lower thresholds like t = 0.05 get a perfect DP and EO of 1.0, but only because everyone is approved, so they don't discriminate at all. The notebook's 'closest to ideal' search lands on that degenerate threshold, which is why I rejected it. The strongest counter-argument is that 0.45 is still unequal on all three measures. Group A is approved at a 14% lower rate (DP = 0.8571), and approvals for Group A are less reliable (PP = 0.8786). A critic could also say that 0.8 is an arbitrary cut-off, that a 6.3-point gain over approving everyone is modest, and that recall on bad credits is only 0.46 at the 0.5 threshold. I accept all of that. The gaps are real, and 0.45 is a compromise, not a fair outcome. Group A also mixes different sexes and marital statuses, and the test sample is small (114 against 186), so these ratios carry sampling error. For those reasons I would only use 0.45 with human review and regular monitoring of the group-level metrics.

• Question 8

If a policymaker had to choose between prioritizing demographic parity or equalized odds for your dataset's specific context, which would you recommend, and why? Cite your own trade-off numbers from Part 2.4. Acknowledge what fairness metric you are sacrificing and why that is acceptable in your view.

- For credit decisions I would prioritise **equalized odds** over demographic parity. A loan decision depends on an outcome, so what matters is that people who would repay get approved equally often, and people who would default get rejected equally often, whichever group they belong to. That is also the easiest rule to explain to an applicant: if you would have repaid, your chances shouldn't depend on who you are. In the training data Group A has a 65.04% good-credit rate and Group B has 72.69%. Forcing equal approval rates would mean either approving more risky Group A applicants or rejecting qualified Group B ones. In my sweep the two metrics actually move together. Going from t = 0.75 to t = 0.45, DP rises from 0.6746 to 0.8571 (+27.05%) and EO rises from 0.4554 to 0.8978 (+97.14%), so choosing EO doesn't cost much DP. The real tension is with predictive parity, which falls from 0.9990 to 0.8786, a 12.05% loss. The trade-off cost is 12.05 / 27.05 = 0.45, which is a cheap price. The metric I am giving up is strict demographic parity. At t = 0.45, Group A's approval rate is still 14.3% below Group B's (DP = 0.8571). I think that is acceptable because part of the gap reflects different repayment rates in the data, not purely discrimination. That is only fair if we keep monitoring the gap, let applicants appeal, and look into why the base rates differ. If the difference comes from historical disadvantage rather than repayment ability, then DP should count for more. That is a values decision for policymakers, and these data can't settle it.

5. Part 4: Reflection

"If a policy-maker asked you: 'just give me one number: should we deploy this model or not?', what would you say, and what is the limitation of that 'one number' answer that you discovered yourself through your analysis?"

- Conditional deployment only. I would use the model as decision support, with a person reviewing each case at a threshold near 0.45, and not as an automatic decision-maker. The main reason is that a ROC-AUC of 0.7952 shows real ranking skill, but no single number shows who the model fails.

- A single number hides a lot, and I ran into at least five problems myself. First, **accuracy hides who is wrong.** An accuracy of 0.7367 sounds fine, but it is only 3.7 points above the 0.70 you would get by approving everyone, and recall for bad credits is just 0.46, so the model misses more than half of the people who will default. Averages also hide groups: the 20 confident errors are concentrated among younger applicants (average age 29.65 against 35.93), and the headline score says nothing about that. Second, **the threshold is a values question, not a fact in the data.** Moving from t = 0.45 to t = 0.75 raises predictive parity from 0.8786 to 0.9990, but equalized odds falls from 0.8978 to 0.4554 and accuracy from 0.7633 to 0.6767. Whether it is worse to lend to someone who will default or to turn away someone who would repay is not something the data can answer. Third, **the fairness metrics conflict.** No threshold satisfies DP, EO and PP together, and PP is best exactly where DP (0.6746) and EO (0.4554) are worst. A single fairness number would quietly pick a winner. Fourth, **confounds stay hidden.** Dropping one variable, checking_status, moved the age coefficient by 22.71% and duration by 11.83%, and variables that aren't in the data could do the same without anyone noticing. Fifth, **subgroup effects are fragile.** The credit-amount coefficient flips sign between personal-status groups (+5.1e-05 against -7.0e-05) while staying statistically insignificant (p = 0.468 and 0.114), and the pooled number (-2.889e-05) hides all of that. Sixth, **a single number can't show its own uncertainty.** With only 300 test cases and 114 applicants in Group A, things like PP = 1.0128 at t = 0.85 or EO bouncing back to 0.5684 at t = 0.95 are partly sampling noise, so even an honest-looking figure would suggest more precision than the data have. In short, one number squashes a multi-dimensional trade-off into false certainty, and the person making the decision should see the dimensions that got squashed.

- These come from my own results: 79 of 300 cases (26.33%) change class depending on which threshold rule is used (Part 1), a 22.71% age shift from omitting one variable (3.2), 20 confident errors with 85% false positives (3.3), and DP 0.8571, EO 0.8978 and PP 0.8786 at t = 0.45 against 0.6746, 0.4554 and 0.9990 at t = 0.75 (3.4).

- The main lesson for me is that responsible deployment means being open about trade-offs, not hiding them behind one metric. A model can be accurate and still unfair to specific people. What I would hand over is a documented threshold, the known failure regions, ongoing monitoring of group-level metrics and an appeal route for every rejected applicant.